

Equations for the elastic Earth

Chapter 2

In this chapter, I summarize the governing equations of an elastic medium. For a deeper understanding, please refer to *Theoretical Global Seismology* by Dahlen and Tromp.⁽¹⁾

§2.1 A microscopic model for Hooke's law

For a better understanding of P- and S-wave propagation in an elastic medium, let us consider a simple mass-spring model. In particular, for S-wave propagation, a **cross spring** is important.

2.1.1 1-D case

Restoring force (stress) against deformation in an elastic medium causes seismic wave propagation. First, let us consider a 1-D case for simplicity. Figure 2.1 shows an example where masses (with mass m) are connected by springs (with spring constant k).

The equations of motion can be written by

$$m \frac{\partial^2 s}{\partial t^2} = k(s_{i+1} - 2s_i + s_{i-1}). \quad (2.1)$$

The right-hand term is the second-order finite difference of s . If the spatial interval of mass Δx is

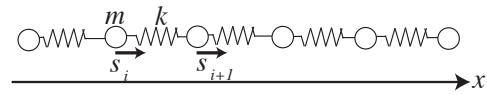


Figure 2.1: A 1-D mass-spring model.

small enough, in the limit of a continuum, the equation becomes a wave equation as,

$$\frac{m}{\Delta x} \frac{\partial^2 s}{\partial t^2} = (k\Delta x) \frac{\partial^2 s}{\partial x^2}, \quad (2.2)$$

where $m/\Delta x$ represents the density ρ , and $k\Delta x$ represents the elastic modulus κ . Here κ satisfies the relation $\kappa = \rho c^2$ between ρ and the wave speed c . The wave equation can be rewritten by

$$\rho \frac{\partial^2 s}{\partial t^2} = \kappa \frac{\partial^2 s}{\partial x^2}. \quad (2.3)$$

Since strain E is given by $E = \frac{\partial s}{\partial x}$, the stress T is expressed as $T = \kappa E$. Therefore, the wave equation

$$\rho \frac{\partial^2 s}{\partial t^2} = \frac{\partial T}{\partial x} \quad (2.4)$$

can be interpreted by the equations of motion of the elastic body. Because the direction of particle motions coincides with the propagation direction, in this case, the wave corresponds to a P wave. Details will be explained in the next chapter.

2.1.2 2-D case: P and S waves

The P wave propagates faster than the S wave. The travel time difference is crucial for locating the hypocenter of an earthquake (e.g., the Omori formula^{note 1)}). An earthquake early warning system forecasts the arrival of the large S wave using the preceding P wave arrivals. Why is the P wave faster than the S wave?

First, to understand the properties of a P wave, let us consider a thought experiment as illustrated in Figure 2.2. Initially, imagine a thin, uniform sheet under uniform pressure from its surroundings, causing it to shrink uniformly. Since a P wave is a wave where compressed and expanded regions propagate alternately, let us align these compressed and decompressed thin sheets adjacent to each other. However, to make this a continuous deformation of a P wave, they must be welded together without any gaps at the boundaries. At these interfaces, additional forces perpendicular to the propagation direction must be applied to resolve the discontinuous deformation. This

boundary correction is exactly the shear deformation that corresponds to an S wave. At first glance, **it might seem that a P wave is composed only of expansion and compression**, but it is crucial to note that it actually involves shear deformation in addition to fluid-like volumetric

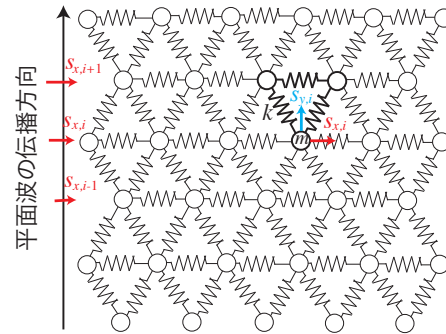
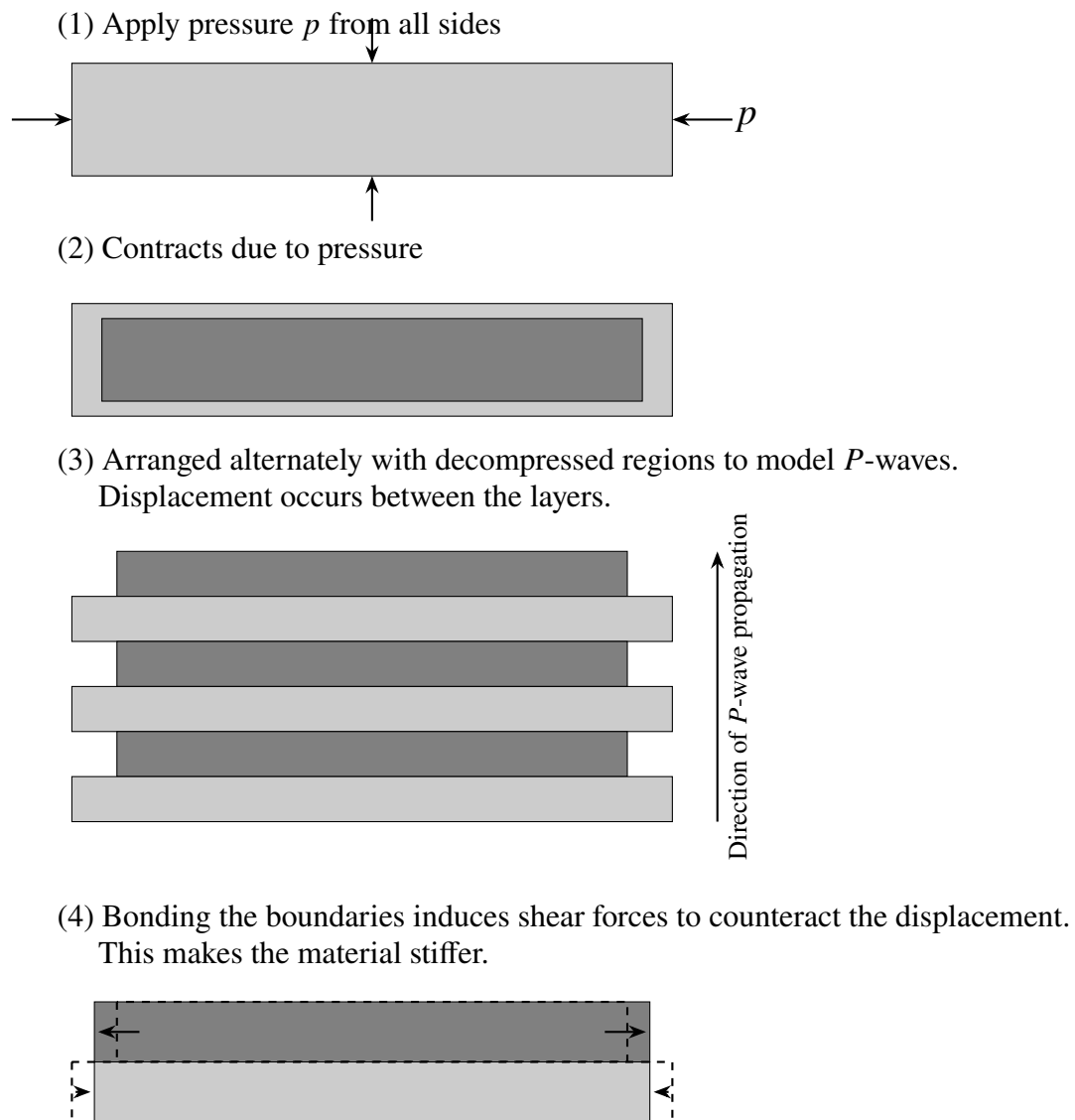


Figure 2.3: A 2-D mass-spring model.

note 1) You can find the paper⁽⁷⁾ at <http://hdl.handle.net/2261/32677>.

Figure 2.2: Deformation of the medium associated with P -wave propagation.

change. The reason a P wave shows a stiffer response and propagates faster than an S wave can be interpreted as this additional rigidity associated with the volumetric change.

Let us extend the 1-D mass-spring model to a 2-D one. **Cross-springs** are crucial for modeling S-wave propagation in 2-D. Here, we consider a simple model as shown in Figure 2.3.

First, let us consider the plane-wave propagation of the S wave in the y direction. The displacement does not depend on x . The i -th mass is moved with a displacement s_{xi} in the x direction. The spring shown by the thick line in the figure exerts a restoring force T^S on the mass m . T^S can be written as,

$$T_i^S = \frac{1}{2}k(s_{x,i+1} - s_{x,i}). \quad (2.5)$$

Because the lower spring also exerts a restoring force on the mass, the total restoring force is $T_i^S - T_{i-1}^S$. Then we obtain a discretized wave equation.

Next, let us consider P-wave propagation. The displacement does not depend on y . The i th mass is moved with a displacement s_{yi} in the y direction. The spring shown by the thick line in the Figure exerts a restoring force T^P on the mass m . T^P in the x direction can be written as,

$$T_i^P = \frac{3}{2}k(s_{y,i+1} - s_{y,i}). \quad (2.6)$$

The total restoring force is $T_i^P - T_{i-1}^P$. A comparison of equation 2.6 with equation 2.5 shows that the restoring force of the P wave is stronger than that of the S wave. This leads to faster P-wave propagation than that of the S-wave.

Last, the relation of the mass-spring model to the Lamé constant is clarified as follows. A model depicted in Figure 2.3 leads to $\lambda = \mu = \sqrt{3}/4k$. In this case, the S-wave β speed is determined by rigidity μ and density ρ , whereas the P-wave velocity V_p is related to both λ and μ as follows:

$$\begin{aligned} \alpha &= \sqrt{\frac{\lambda + 2\mu}{\rho}} \\ \beta &= \sqrt{\frac{\mu}{\rho}}. \end{aligned} \quad (2.7)$$

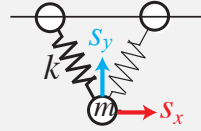
For P-wave propagation, λ represents the restoring force related to fluid pressure, whereas μ represents the restoring force to suppress tangential motions.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

Problem 2.1

Derive equation 2.5 and equation 2.6.

Hint: Because we consider a plane wave, relative motions depending on x are negligible. For the simple mode shown in the right Figure, calculate the restoring forces for displacements s_x and s_y .



2.1.3 [†] Attenuation

Although most of this lecture note does not cover seismic attenuation, the observed seismic waves decay with time. This is because elastic energy is gradually dissipated into thermal energy. This subsection briefly summarizes physical models of seismic attenuation in a spring–mass system. For simplicity, we will consider connecting a damper, which is responsible for viscous dissipation, to the spring–mass system considered in the previous section.

Dashpot

If only the spring is considered, there is no energy dissipation (i.e., no seismic attenuation). To account for damping due to energy dissipation, a dashpot is added to the spring.

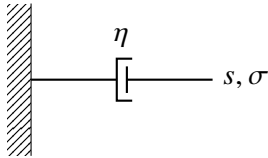


Figure 2.4: Dashpot

Here we consider a displacement $s(t)$ given at the lower right end. Since the dashpot is under viscous resistance η , the applied force $\sigma(t)$ is given by

$$\sigma = \eta \frac{ds}{dt}. \quad (2.8)$$

Kelvin–Voigt model

Here, we consider a model of a spring–mass system with a dashpot.

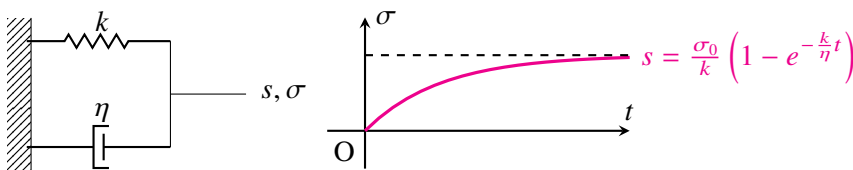


Figure 2.5: Kelvin–Voigt model and the response.

In this case, the relationship between the applied force σ at the lower right end and the displacement s can be written as:

$$\sigma = k \left(s + \frac{\eta}{k} \frac{ds}{dt} \right). \quad (2.9)$$

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On the right-hand side, when a constant force σ_0 is applied at time t , the force σ is given by:

$$\begin{cases} 0, & t < 0, \\ \sigma_0, & t \geq 0. \end{cases} \quad (2.10)$$

The displacement (Creep response function $J(t)$) when a constant force σ_0 is applied at time t is:

$$s = \frac{\sigma_0}{k} \left(1 - e^{-\frac{k}{\eta}t} \right). \quad (2.11)$$

If only a spring is considered, the displacement would be σ_0/k instantaneously (black dashed line), but due to viscous relaxation, it can be observed that the displacement reaches the final value with a delay.

Zener model

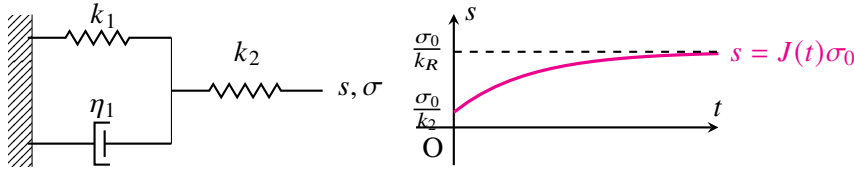


Figure 2.6: Zener model and the response.

Let us consider a more realistic model by adding one more spring.

The relationship between the force σ applied at the lower right edge and the displacement s is represented by

$$\sigma(t) + \tau_\epsilon \frac{d\sigma}{dt} = k_R \left(s + \tau_\sigma \frac{ds}{dt} \right), \quad (2.12)$$

where k_R , τ_ϵ , and τ_σ are defined as

$$k_R \equiv \frac{k_1 k_2}{k_1 + k_2}, \tau_\sigma \equiv \eta/k_1, \tau_\epsilon \equiv \eta/(k_1 + k_2). \quad (2.13)$$

Similarly, if we apply a force $\sigma = \sigma_0 H(t)$ ($H(t)$ is the Heaviside function) at the right end. If a force is applied, the response is $\sigma_0 J(t)$, where $J(t)$ is the creep response function given by

$$J(t) = \frac{1}{k_R} \frac{\tau_\epsilon}{\tau_\sigma} + \frac{1}{k_R} \frac{\tau_\sigma - \tau_\epsilon}{\tau_\sigma} \left[1 - e^{-\frac{t}{\tau_\sigma}} \right]. \quad (2.14)$$

Complex elastic constant

Here, we consider a periodic input σ as $\sigma = \sigma_0 e^{-i\omega t}$, $s = s_0 e^{-i\omega t}$. After the substitution, the Kelvin-Voigt model is given by

$$\sigma_0 = k \left(1 - i\omega \frac{\eta}{k} \right) s_0, \quad (2.15)$$

and the Zener model is given by

$$\sigma_0 = k_R \frac{1 - i\omega\tau_\sigma}{1 - i\omega\tau_\epsilon} s_0. \quad (2.16)$$

With the complex elastic constant K , they can be rewritten as follows:

$$K_v \equiv k \left(1 - i\omega \frac{\eta}{k} \right) \quad (2.17)$$

$$K_z \equiv k_R \frac{1 - i\omega\tau_\sigma}{1 - i\omega\tau_\epsilon}. \quad (2.18)$$

Thus, by extending the elastic constants to complex numbers, the attenuation can be described.

In the case of the Zener model, Q^{-1} is given by

$$Q^{-1}(\omega) = \frac{\tau_\sigma - \tau_\epsilon}{\tau} \frac{\omega\tau}{1 + \omega^2\tau^2}, \quad (2.19)$$

where $\tau \equiv \sqrt{\tau_\sigma\tau_\epsilon}$.

Anelastic parameter Q

Let us consider displacement s

$$s(t) = s_2 \cos(\omega t - \delta) \quad (2.20)$$

for a given input σ

$$\sigma(t) = \sigma_2 \cos \omega t, \quad (2.21)$$

where σ_2 , s_2 and ω are real constants.

When the energy E decreases by ΔE during one period of oscillation, the anelastic parameter Q is

$$\frac{2\pi}{Q} = \frac{\Delta E}{E} \quad (2.22)$$

This can be expressed as ΔE , the amount of energy dissipated in one cycle, is

$$\Delta E = \int_0^{\frac{2\pi}{\omega}} \sigma \frac{ds}{dt} dt, \quad (2.23)$$

and E is

$$E = \frac{1}{2} \sigma(0) s(0) \quad (2.24)$$

can be evaluated as

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Physical dispersion

To consider the dispersion of phase velocity caused by damping, we generalize the spring-mass model. Here, we consider the Zener model instead of the spring and similarly consider periodic oscillations characterized by the angular frequency ω . In this case, by taking Δx to be small, the wave equation with coefficients extended to complex numbers can be written as follows:

$$-\omega^2 \rho s_0(x) = K_v \frac{\partial^2 s_0(x)}{\partial x^2}. \quad (2.25)$$

Let us consider displacement s_0 , which describes wave propagation with phase velocity $c(\omega)$ as

$$s_0(x) = e^{-\frac{\omega x}{2c(\omega)Q(\omega)}} e^{i\omega \frac{x}{c(\omega)}}. \quad (2.26)$$

In this case, $c(\omega)$ changes with frequencies (dispersion). The dispersion relation can be written as

$$c(\omega) = \sqrt{\frac{k_R}{\rho}} \left(1 + \frac{\tau_\sigma - \tau_\epsilon}{2\tau} \frac{\omega^2 \tau^2}{1 + \omega^2 \tau^2} \right). \quad (2.27)$$

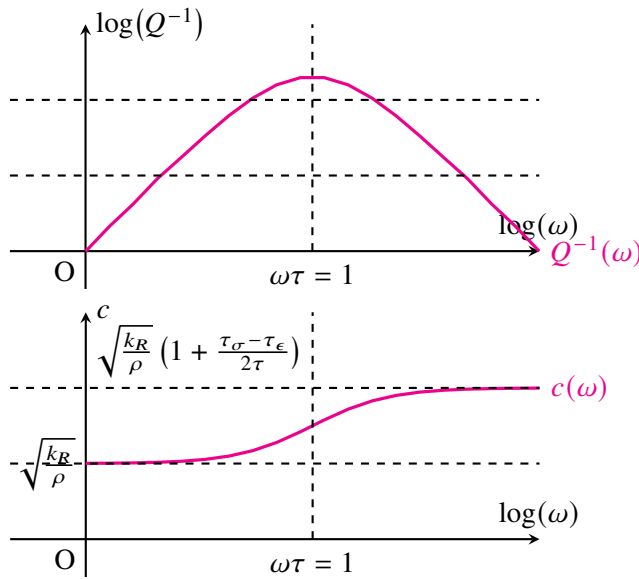


Figure 2.7: Physical dispersion

In the seismic frequency band (10^{-3} – 10 Hz), Q is known to be approximately constant and frequency-independent. Combining a spring-mass system with a dashpot is known to explain the frequency independence of Q . Now, in the Zener model, corresponding to a single relaxation time (τ_σ), there is a frequency band in which the damping works, but the phase speed changes significantly with frequency. In reality, the Earth has many different relaxation time scales associated

with various physical processes. Therefore, by combining Zener models with different relaxation times, we can extend the frequency range over which Q^{-1} remains constant.

Problem 2.2

1. When a wave propagates through a medium, the medium dissipates the oscillation energy, converting it into heat, which causes the wave to attenuate. The propagation of a wave s_0 oscillating at angular frequency ω can be expressed using the anelastic parameter Q as:

$$s_0(x, t) = e^{-\frac{|\omega|x}{2cQ}} e^{i\omega(\frac{x}{c}-t)} \quad (2.28)$$

where Q is a positive real number. Consider the wave generated when an impulse is applied at $x = 0$ at $t = 0$. Since the impulse contains all frequency components equally, the wave packet propagating in the positive x direction for $t > 0$ and $x > 0$ can be written as:

$$s(x, t) = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-\frac{|\omega|x}{2cQ}} e^{i\omega(\frac{x}{c}-t)} d\omega \quad (2.29)$$

Assuming c and Q are constant with respect to frequency, find the analytical solution for s . Also, illustrate the time evolution at position x .

2. The solution obtained in (1) is, strictly speaking, physically impossible. Discuss the reasons for this.

§2.2 Lagrangian and Eulerian variables

Here we consider a particle at $\mathbf{x}$ in a continuum at time 0. ^{note 2)} It moves with time as $\mathbf{r}(\mathbf{x}, t)$. A Lagrangian variable describes a quantity over time by its initial location $\mathbf{x}$, and an Eulerian variable describes a quantity over time in a fixed frame as $\mathbf{x}(\mathbf{r}, t)$. An Eulerian quantity q^E can be related to the Lagrangian quantity q^L as,

$$q^L(\mathbf{x}, t) = q^E(\mathbf{r}(\mathbf{x}, t), t). \quad (2.30)$$

Time derivatives of the equation lead to

$$\partial_t q^L = \partial_t q^E + \mathbf{u}^E \cdot \nabla_{\mathbf{r}} q^E \equiv D_t q^E, \quad (2.31)$$

where $\mathbf{u}^E$ is Euler velocity. Material derivative D_t represents the Lagrangian time derivative of a particle in the Eulerian form. Later, ∂_t is an abbreviation of $\frac{\partial}{\partial t}$. This relation holds for vector quantities as particle velocity $\mathbf{u}^L$ and $\mathbf{u}^E$. As a q^E , here we consider acceleration $\partial_t \mathbf{u}^L$ as

$$\partial_t \mathbf{u}^L = \partial_t \mathbf{u}^E + \mathbf{u}^E \cdot \nabla_{\mathbf{r}} \mathbf{u}^E \equiv D_t \mathbf{u}^E, \quad (2.32)$$

When we consider seismic records, the Lagrangian description is more natural because the seismometer is pinned at a surface point.

Later, we consider an infinitesimal deformation in a framework of linear elasticity. Let us consider small deformation $\mathbf{s}$ as,

$$\mathbf{r}(\mathbf{x}, t) = \mathbf{x} + \mathbf{s}(\mathbf{x}, t). \quad (2.33)$$

$$\begin{aligned} q^E(\mathbf{r}, t) &= q_0(\mathbf{r}) + q^{E1}(\mathbf{r}, t), \\ q^L(\mathbf{x}, t) &= q_0(\mathbf{x}) + q^{L1}(\mathbf{x}, t). \end{aligned} \quad (2.34)$$

Because $\mathbf{s}$ is small, the perturbations q^{E1} and q^{L1} of Eulerian variable and Lagrangian one can be written by,

$$q^{E1}(\mathbf{r}, t) = q^{E1}(\mathbf{x}, t), q^{L1}(\mathbf{x}, t) = q^{L1}(\mathbf{r}, t). \quad (2.35)$$

We note that the perturbations do not depend on $\mathbf{x}, \mathbf{r}$.

Next, let us consider the relationship between an Eulerian perturbation q^{E1} and a Lagrangian one q^{L1} . When we consider first-order perturbations, q^{L1} can be written as ^{note 3)}

$$q^{L1}(\mathbf{x}, t) = q^{E1}(\mathbf{r}, t) + \mathbf{s} \cdot \nabla_{\mathbf{r}} q_0. \quad (2.36)$$

This equation is the integration of material derivatives. In a framework of linear elasticity, because we can neglect the second term on the right-hand side, we do not need to distinguish Eulerian from Lagrangian. However, because stress and density have initial values (∇q_0), the spatial derivatives cause the discrepancy between the Eulerian and Lagrangian descriptions. When we must consider initial stress (e.g., hydrostatic pressure due to gravity), we need to be aware of the distinction between the descriptions. In §??, we will take such an example.

^{note 2)} A bold symbol like $\mathbf{x}$ denotes a vector in this textbook.

^{note 3)} See §3.2 of Dahlen and Tromp (1998)⁽¹⁾ for details.

§2.3 Strain

In order to measure the deformation of an elastic body, let us trace two particles ($\mathbf{r}, \mathbf{r} + d\mathbf{r}$). At $t = 0$, they are located at $\mathbf{x}, \mathbf{x} + d\mathbf{x}$. $d\mathbf{r}$ can be related to $d\mathbf{x}$ as,

$$d\mathbf{r} = \nabla_{\mathbf{x}} \mathbf{r} \cdot d\mathbf{x} = (\mathbf{I} + \nabla_{\mathbf{x}} \mathbf{s}) \cdot d\mathbf{x}. \quad (2.37)$$

The change of the distance between the particles is estimated to be

$$|d\mathbf{r}|^2 - |d\mathbf{x}|^2 = 2(d\mathbf{x} \cdot \mathbf{E}^L \cdot d\mathbf{x}). \quad (2.38)$$

Here $\mathbf{E}^L$ is the Green-Lagrange strain defined by

$$\mathbf{E}^L \equiv \frac{1}{2} [\nabla \mathbf{s} + (\nabla \mathbf{s})^T - (\nabla \mathbf{s})^T \nabla \mathbf{s}] = \frac{1}{2} \left(\frac{\partial s_j}{\partial x_i} + \frac{\partial s_i}{\partial x_j} + \frac{\partial s_l}{\partial x_i} \frac{\partial s_l}{\partial x_j} \right). \quad (2.39)$$

When $\mathbf{s}$ is small enough to neglect the second-order term, the strain can be simplified to

$$\mathbf{E}^L = \frac{1}{2} [\nabla \mathbf{s} + (\nabla \mathbf{s})^T] = \frac{1}{2} \left(\frac{\partial s_j}{\partial x_i} + \frac{\partial s_i}{\partial x_j} \right) \quad (2.40)$$

Here, following Einstein's summation convention, we sum over all values of the index. Although the strain tensor $\mathbf{E}^L$ is a Lagrangian variable, we do not need to distinguish it from its Eulerian counterpart for infinitesimal deformations. The strain tensor has 6 independent components because of the symmetry.

Problem 2.3

Calculate the strain tensor up to the first order when considering a rigid rotation with an infinitesimal angle in 2-D. You can find second-order terms, although rigid rotation should not, in principle, cause strain. Next, show that the Green-Lagrange strain of an infinitesimal rigid rotation vanishes completely.

2.3.1 Strain in an arbitrary coordinate

Here we consider linear strain in an arbitrary coordinate. A displacement vector $\mathbf{s}$ can be represented by orthogonal unit vectors as

$$\mathbf{s} = \sum_j s_j \hat{\mathbf{x}}_j. \quad (2.41)$$

The gradient of the vector is expressed by

$$\nabla \equiv \hat{\mathbf{x}}_i \frac{\partial}{\partial x_i}. \quad (2.42)$$

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The insertion of the definition of s leads to the following equation:

$$\nabla s = \sum_i \hat{x}_i \frac{\partial s}{\partial x_i} = \sum_i \hat{x}_i \left[\sum_j \frac{\partial s_j}{\partial x_i} \hat{x}_j + \sum_j s_j \frac{\partial \hat{x}_j}{\partial x_i} \right]. \quad (2.43)$$

The second-order tensor E can be represented by basis vectors and the corresponding components as

$$E = \sum_{ij} E_{ij} \hat{x}_i \hat{x}_j. \quad (2.44)$$

The complication originates from the partial derivatives of the basis vectors.

Strain in a cylindrical coordinate (r, φ, z)

Here we consider strain in a cylindrical coordinate (r, φ, z) . Partial derivatives of the unit vectors $\hat{r}, \hat{\varphi}, \hat{z}$ are given by:

$$\begin{aligned} \frac{\partial \hat{r}}{\partial r} &= 0, & \frac{\partial \hat{r}}{\partial \varphi} &= \hat{\varphi}, & \frac{\partial \hat{r}}{\partial z} &= 0 \\ \frac{\partial \hat{\varphi}}{\partial r} &= 0, & \frac{\partial \hat{\varphi}}{\partial \varphi} &= -\hat{r}, & \frac{\partial \hat{\varphi}}{\partial z} &= 0 \\ \frac{\partial \hat{z}}{\partial r} &= 0, & \frac{\partial \hat{z}}{\partial \varphi} &= 0, & \frac{\partial \hat{z}}{\partial z} &= 0. \end{aligned} \quad (2.45)$$

The definition of the strain with the above equations leads to the following representation of the corresponding components as:

$$\begin{aligned} E_{rr} &= \frac{\partial s_r}{\partial r}, E_{\varphi\varphi} = \frac{1}{r} \frac{\partial s_\varphi}{\partial \varphi} + \frac{s_r}{r}, E_{zz} = \frac{\partial s_z}{\partial z} \\ 2E_{\varphi z} &= \frac{\partial s_z}{\partial \varphi} + \frac{\partial s_\varphi}{\partial z}, E_{zr} = \frac{\partial s_r}{\partial z} + \frac{\partial s_z}{\partial r}, \\ 2E_{r\varphi} &= r \frac{\partial}{\partial r} \left(\frac{s_\varphi}{r} \right) + \frac{1}{r} \frac{\partial s_r}{\partial \varphi}. \end{aligned} \quad (2.46)$$

Problem 2.4

Derive the strain in a cylindrical coordinate shown above.

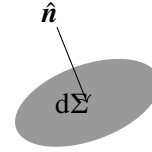
Strain in a spherical coordinate (r, θ, φ)

Here only the results are shown as,

$$\begin{aligned}
 E_{rr} &= \frac{\partial s_r}{\partial r}, E_{\theta\theta} = \frac{1}{r} \frac{\partial s_\theta}{\partial \theta} + \frac{s_r}{r}, E_{\varphi\varphi} = \frac{1}{r \sin \theta} \left(\frac{\partial s_\varphi}{\partial \varphi} + s_\theta \cos \theta \right) + \frac{s_r}{r} \\
 2E_{\theta\varphi} &= \frac{1}{r} \frac{\partial s_\varphi}{\partial \theta} + \frac{1}{r \sin \theta} \left(\frac{\partial s_\theta}{\partial \varphi} - s_\varphi \cos \theta \right), 2E_{\varphi r} = \frac{1}{r \sin \theta} \frac{\partial s_r}{\partial \varphi} + r \frac{\partial}{\partial r} \left(\frac{s_\varphi}{r} \right) \\
 2E_{r\theta} &= r \frac{\partial}{\partial r} \left(\frac{s_\theta}{r} \right) + \frac{1}{r} \frac{\partial s_r}{\partial \theta}.
 \end{aligned} \tag{2.47}$$

§2.4 Stress and traction

Let us consider a small surface $d\Sigma$ in a continuum with the normal vector $\hat{n}$. Traction is defined by force $\mathbf{f}$ per unit area acting on $d\Sigma$. The traction is parallel to the normal vector $\hat{n}$, and the positive sign is defined by force from the positive side according to the normal vector to the negative side. Stress $\mathbf{T}^E$ ^{note 4)} is defined by



$$\mathbf{f} = d\Sigma \hat{n} \cdot \mathbf{T}^E. \tag{2.48}$$

Traction f_{12} of medium 1 from medium 2 is f_{21} .

§2.5 Conservation of angular momentum

Conservation of angular momentum requires that the stress tensor is symmetric as $T_{ij} = T_{ji}$.^{note 5)} as : $T_{ij} = T_{ji}$

This symmetry is held without microscopic spin interaction.⁽³⁾

^{note 4)}This stress is Cauchy stress in a precise manner. There are two other definitions (see Dahlen and Tromp 1998 in detail).

^{note 5)}This symmetry is derived from the equilibrium of the moment of an infinitesimal volume.

Problem 2.5

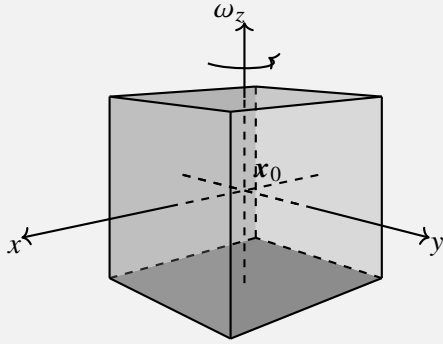
Let us consider angular momentum along the z axis of an infinitesimal cube with a side of ϵ . Show $T_{xy} = T_{yx}$ from the conservation of angular momentum in cases:

1. T_{xy} and T_{yx} are constant, and other T_{ij} are zero.
2. (Optional)

$$T_{xy}(\mathbf{x}) = T_{xy}(\mathbf{x}_0) + T_{xy,x}(\mathbf{x}_0)\delta x + T_{xy,y}(\mathbf{x}_0)\delta y, \quad (2.49)$$

$$T_{yx}(\mathbf{x}) = T_{yx}(\mathbf{x}_0) + T_{yx,x}(\mathbf{x}_0)\delta x + T_{yx,y}(\mathbf{x}_0)\delta y, \quad (2.50)$$

and other $T_{ij} = 0$.



§2.6 Conservation of mass

Let us consider the governing equations, conservation of mass first. This equation becomes important for the buoyancy force.

In general, conservation of mass can be written as,

$$\partial_t \rho^E = -\nabla \cdot (\rho^E \mathbf{u}^E), \quad (2.51)$$

where $\mathbf{u}^E$ is the Eulerian particle velocity.

Here, we consider the first-order perturbation of the density as

$$\rho^E = \rho_0 + \rho^{E1}. \quad (2.52)$$

Time integration of the first-order perturbation leads to

$$\rho^{E1} = -\nabla \cdot (\rho_0 \mathbf{s}). \quad (2.53)$$

The right-hand term represents the divergence of the mass flux, whereas the left one represents the change in mass. On the other hand, the Lagrangian form can be written as

$$\rho^{L1} = -\rho_0 \nabla \cdot (\mathbf{s}). \quad (2.54)$$

Because we focused on a particle, the right-hand side shows the corresponding expansion or deflation.

We note the discrepancy that the Lagrangian density perturbation ρ^{L1} is $\rho^{L1} = -\rho_0 \nabla \cdot \mathbf{s}$. If $\nabla \rho_0$ is 0, they are the same. However, they are different in general. When the wavelength of a seismic wave is much smaller than the typical scale of density change, we can neglect the difference.

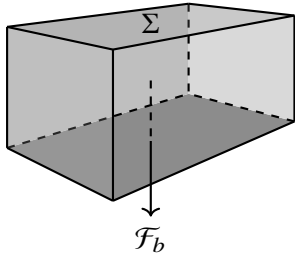
§2.7 Equations of motion: conservation of momentum

Let us consider a temporal change in a volume V^t .

$$\frac{d}{dt} \int_{V^t} \rho^E \mathbf{u}^E dV^t = \mathcal{F}, \quad (2.55)$$

The external force $\mathbf{F}$ can be described by the surface force acting on ∂V^t and the body force acting throughout the volume, such as gravity and electromagnetic force as

$$\mathcal{F} = \mathcal{F}_s + \mathcal{F}_b = \int_{\partial V^t} (\hat{\mathbf{n}}^t \cdot \mathbf{T}^E) d\Sigma^t + \int_{V^t} \rho^E \mathbf{g}^E dV^t. \quad (2.56)$$



With the help of Gauss's divergence theorem, they can be written as

$$\rho^E D_t \mathbf{u}^E = \nabla_r \cdot \mathbf{T}^E - \rho^E \nabla \phi^{E1}. \quad (2.57)$$

If we can neglect initial stress up to the first order, it can be simplified as

$$\rho_0 \partial_t^2 \mathbf{s} = \nabla \cdot \mathbf{T}^{E1}. \quad (2.58)$$

Hydrostatic equilibrium

When we consider hydrostatic pressure, the equations of motion up to the first order can be written as

$$\rho_0 \partial_t^2 \mathbf{s} = \nabla \cdot \mathbf{T}^E - \rho^E \nabla \phi^E, \quad (2.59)$$

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

where ϕ^E is the gravitational potential. ^{note 6)} When we consider gravity, we must consider hydrostatic pressure, which sustains the gravity force. Deviatoric stress T^{E1} from the hydrostatic pressure can be expressed as follows:

$$T_{ij}^E = -p_0 \delta_{ij} + T_{ij}^{E1}. \quad (2.60)$$

Because the hydrostatic pressure sustains the gravity force, the pressure p_0 should satisfy the relation as follows,

$$\nabla p_0 = -\rho_0 \nabla \phi_0 = \rho_0 \mathbf{g}. \quad (2.61)$$

They lead to

$$\rho_0 \partial_t^2 \mathbf{s} = \nabla \cdot \mathbf{T}^{E1} - \nabla [\rho_0 \mathbf{s} \cdot \nabla \phi_0] - \rho_0 \nabla \phi^{E1} - \rho^{E1} \nabla \phi_0. \quad (2.62)$$

Additionally, since the perturbation of the gravitational potential ϕ^{E1} is caused by the density perturbation ρ^{E1} , it must satisfy Poisson's equation:

$$\nabla^2 \phi^{E1} = 4\pi G \rho^{E1}. \quad (2.63)$$

§2.8 Conservation of energy

Total energy (kinetic energy + elastic energy) U , and energy flux $\mathbf{K}$ can be expressed as follows:

$$U = \frac{1}{2} \left[\rho_0 \partial_t \mathbf{s} \cdot \partial_t \mathbf{s} + \sum_{ij} E_{ij} T_{ij} \right], \quad (2.64)$$

$$\mathbf{K} = -\mathbf{T} \cdot \partial_t \mathbf{s}. \quad (2.65)$$

Conservation of energy can be written as,

$$\partial_t U + \nabla \cdot \mathbf{K} = 0. \quad (2.66)$$

Elastic energy W can be defined by

$$W = \frac{1}{2} \sum_{ij} E_{ij} T_{ij}. \quad (2.67)$$

Stress can be represented by the spatial gradient of W as

$$T_{ij} = \frac{\partial W}{\partial E_{ij}}. \quad (2.68)$$

^{note 6)} Here, we consider gravity, but we neglect perturbation of gravity (an effect of self-gravitation). This approximation is known as the Cowling approximation in astrophysics. In particular, this approximation is effective for a stratified atmosphere. The effect of self-gravitation becomes important in a period longer than 3000 s, as described later.

We will discuss the condition for the existence of elastic energy W as a potential energy below.

Work per unit volume δR done by the internal stress T_{ij} can be given⁽⁴⁾ by

$$\delta R = -T_{ij}\delta E_{ij}. \quad (2.69)$$

For deformation from $\mathbf{E} = 0$ to $\mathbf{E} = \Delta\mathbf{E}$ of a given infinitesimal volume, the corresponding work ΔR by the internal stress is given by

$$\Delta R = - \int_0^{\Delta\mathbf{E}} \sum_{ij} T_{ij} dE_{ij}, \quad (2.70)$$

which depends on the deformation history. Then we deform it again from $\mathbf{E} = \Delta\mathbf{E}$ to $\mathbf{E} = 0$. Elastic deformation requires $\Delta R = 0$ because the internal stress should be conservative. The conservative property requires the following condition

$$\frac{\partial T_{ij}}{\partial E_{kl}} = \frac{\partial T_{kl}}{\partial E_{ij}}. \quad (2.71)$$

In the 2-D case (in this case, the number of the independent stress/strain components is 3), this condition can be interpreted as vortex-free (Problem 2.8). Consequently, the elastic energy W can be interpreted as a scalar potential.

Problem 2.6

Derive

$$dR = -T_{ij}dE_{ij}. \quad (2.72)$$

(Hint: Estimate work done by the internal force $F_j = \partial_i T_{ij}$.)

Problem 2.7

(1)

Let us consider the works for two different deformation paths for a simple 2-D case. A rectangular area is deformed according to the strain of $E_{xx} = \Delta E_{xx}$, $E_{yy} = \Delta E_{yy}$ with two different paths: (i) first deform the area with ΔE_{yy} , then deform it with ΔE_{xx} . (ii) First deform it with ΔE_{xx} , then deform it with ΔE_{yy} as shown in Figure 2.8. Estimate works of $\Delta R^{(i)}$ and $\Delta R^{(ii)}$ for the cases (i) and (ii), respectively.

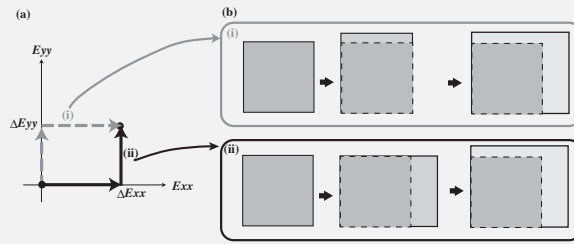


Figure 2.8:

When the stress satisfies the condition of a conservative force, $\Delta R^{(i)} = \Delta R^{(ii)}$ is required. Show the following condition of elastic modulus, which satisfies the above requirement.

$$C_{ijkl} = C_{klij}, \quad (2.73)$$

Problem 2.8

In order to understand the relation (equation 2.71):

$$\frac{\partial T_{ij}}{\partial E_{kl}} = \frac{\partial T_{kl}}{\partial E_{ij}}, \quad (2.74)$$

let us consider the 2-D case (3 independent variables of stress/strain components). Derive the condition for the case that works δR , which is work done by the internal stress, does not depend on deformation paths.

This condition guarantees that the spatial gradient of a scalar potential W can represent stress $\mathbf{T}$. Thus elastic energy W can be interpreted as the scalar potential for stress $\mathbf{T}$.

§2.9 Constitutive equation: Hooke's law

To determine elastic deformation, we must know the constitutional relation between stress T^{L1} and strain E_{L1} . For understanding the deformation of an arbitrary volume, a Lagrangian description is essential. First, we do not account for the difference between the Lagrangian and Eulerian formulations without considering the initial stress. For a linear elastic medium, a relation between stress and strain can be represented by Hooke's law as

$$T_{ij} = C_{ijkl}E_{kl}, \quad (2.75)$$

where C_{ijkl} is the elastic tensor with 81 components. The symmetry of stress and strain tensors leads to the symmetry of C_{ijkl} as $C_{ijkl} = C_{jikl}$, $C_{ijkl} = C_{ijlk}$. The 1st law of thermodynamics requires $C_{ijkl} = C_{klij}$ ^{note 7)}. As a result, the elastic tensor has 21 independent components.

2.9.1 Betti–Rayleigh reciprocity theorem

We introduce the Betti–Rayleigh reciprocity theorem for the representation theorem of an elastic medium, which is explained later.

Let us consider a situation where forces f_1 and f_2 are applied. We assume that dynamic forces can be neglected. We evaluate the work done in two cases: applying f_1 followed by f_2 , and applying f_2 followed by f_1 (Figure 2.9). Here, the forces are applied quasi-statically. First, f_1 is gradually applied from 0 to f_1 . After reaching a steady state, f_2 is gradually applied from 0 to f_2 . The other deformation path reverses the order of the applied forces. As considered in Problem 2.7, the work done by internal forces is independent of the deformation path (using Maxwell's reciprocity theorem). Therefore, only the work done by traction and body forces contributes.

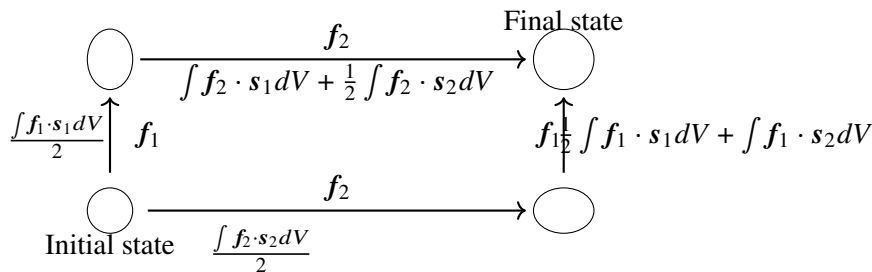


Figure 2.9: Two different deformation paths.

^{note 7)} See a textbook on continuum mechanics for details.

Since the work done in both deformation paths is equal, we can derive the following relation:

$$\int_V \mathbf{f}_1(\mathbf{x}) \cdot \mathbf{s}_2 dV - \int_V \mathbf{f}_2(\mathbf{x}) \cdot \mathbf{s}_1 dV = - \int_{\Sigma} [\mathbf{T}_1 \cdot \mathbf{s}_2(\mathbf{x})] \cdot \hat{\mathbf{n}} d\Sigma + \int_{\Sigma} [\mathbf{T}_2 \cdot \mathbf{s}_1(\mathbf{x})] \cdot \hat{\mathbf{n}} d\Sigma. \quad (2.76)$$

This is known as the Betti–Rayleigh reciprocity theorem.

By substituting $\mathbf{f}_i = \sum_j f_j^i \mathbf{n}_j \delta(\mathbf{x} - \mathbf{x}_j)$ and defining the corresponding displacements at each point as $s_j \mathbf{n}_j$ (where $\mathbf{x}_j$ is a unit vector pointing in an arbitrary direction), and assuming no traction (the right-hand side is 0), we can derive Betti's theorem, well known in solid mechanics:

$$\sum_j f_j^1 s_j^2 = \sum_j f_j^2 s_j^1. \quad (2.77)$$

2.9.2 Isotropic medium

When an elastic medium is isotropic, the elastic tensor can be simplified using Lamé constants λ, μ as,

$$C_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}). \quad (2.78)$$

Here we derive Hooke's law for the isotropic medium explicitly for further sections.

$$\begin{aligned} T_{xx} &= (\lambda + 2\mu)E_{xx} + \lambda(E_{yy} + E_{zz}), & T_{xy} &= 2\mu E_{xy}, & T_{xz} &= 2\mu E_{xz} \\ T_{yy} &= (\lambda + 2\mu)E_{yy} + \lambda(E_{xx} + E_{zz}), & T_{yz} &= 2\mu E_{yz} \\ T_{zz} &= (\lambda + 2\mu)E_{zz} + \lambda(E_{xx} + E_{yy}). \end{aligned} \quad (2.79)$$

There are several different definitions of the elastic constant, although they are identical in theory. Young modulus E and Poisson's ratio ν are also major, and they can be related to Lamé's constant as

$$E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}, \nu = \frac{\lambda}{2(\lambda + \mu)}. \quad (2.80)$$

Here, we consider the effects of hydrostatic pressure on an elastic medium. In the deep Earth, the hydrostatic pressure reaches several hundred GPa; the initial pressure is not negligible in some cases. In order to trace the temporal change of an infinitesimal volume V^t , the variables are described by the Lagrangian as,

$$p^L = p_0 + p^{L1}. \quad (2.81)$$

The constitutional relation can be described by

$$p^{L1} = -\lambda(\mathbf{x}) \nabla \cdot \mathbf{s}. \quad (2.82)$$

Eulerian equations of motion can be written as

$$p^{E1} = p^{L1} - \mathbf{s} \cdot \nabla p_0. \quad (2.83)$$

§2.10 Relation between density and seismic wave velocity

At a planetary scale, the possible values for seismic wave velocities are theoretically bounded. Counterintuitively, we can roughly estimate seismic wave velocities by assuming hydrostatic equilibrium. This provides crucial knowledge for considering seismic wave propagation on other planets.

2.10.1 Sustaining hydrostatic equilibrium with elasticity

Let us estimate the typical seismic wave velocity when considering a global scale. We consider the elastic constant K . Letting the seismic wave velocity be c_s , and the density be ρ , K is given by $k = \rho c_s^2$. Since K has the dimension of pressure, we consider the typical pressure. Letting the Earth's radius be R_e , and the gravitational acceleration at the surface be g , the typical hydrostatic pressure is $\rho g R_e$. Therefore, from dimensional analysis:

$$\rho(c_s)^2 \sim \rho g R_e, \quad (2.84)$$

we can estimate the seismic wave velocity as $c_s \sim \sqrt{g R_e}$. Note that we can estimate the seismic wave velocity solely from Earth's radius and the gravitational acceleration. In this case, the seismic wave velocity is around 8 km/s, which is a surprisingly good estimate.

Let us now consider a state in which the hydrostatic pressure is comparable to the bulk modulus K . Here, the bulk modulus K is defined as

$$K = \rho \left(\alpha^2 - \frac{4}{3} \beta^2 \right) = \rho c_s^2, \quad (2.85)$$

and c_s is called the bulk sound velocity^{note 8)}. If the hydrostatic pressure matches K , the strain E becomes on the order of 1, and for a linear elastic medium, the volume would become 0. In other words, when hydrostatic pressure P is sustained by elasticity, the elastic constant K is required to be greater than or equal to P .

Since hydrostatic pressure is 0 in shallow regions, a naive application of dimensional analysis yields $c_s = 0$. In reality, this is not the case, meaning rocks have an intrinsic rigidity that exceeds what is necessary to support the hydrostatic pressure. Consider isotropically compressing a rock at the surface. Microscopically, the interatomic distances gradually decrease. As a result, the density increases, and the interatomic forces strengthen, which is expected to correspond to increased seismic wave velocities.

Thinking in terms of a planetary average, this particular argument feels slightly contrived. Let us consider the balance between the gravitational potential and accumulated elastic energy^{note 9)}.

note 8) $\phi \equiv c_s^2$ and is also known as the seismic parameter.

note 9) To be precise, the virial theorem leads to the equipartition of gravitational potential and elastic energy under hydrostatic equilibrium.

Assuming a uniform sphere for simplicity, letting U be the total elastic energy and Ω be the total gravitational potential, we obtain:

$$U = 3 \int P dV \sim 3M(c_s)^2 \quad (2.86)$$

$$\Omega = \frac{3}{5}gR_e, \quad (2.87)$$

which leads to $(c_s)^2 = gR_e/5$ and halves the value of c_s ^{note 10)}. This discrepancy arises because calculating the average includes the unrealistic condition that pressure is 0 near the surface ($c_s = 0$). As a result, heuristically taking R_e as the length scale empirically yields an estimate closer to reality. A proper evaluation naturally yields consistent results. Below, we discuss this more quantitatively.

2.10.2 Adams–Williamson equation

Let us consider a situation where hydrostatic pressure P is sustained by adiabatic compression. The pressure gradient must balance gravity:

$$\frac{dP}{dr} = -\rho g(r). \quad (2.88)$$

For a spherically symmetric structure, the gravitational acceleration g is

$$g(r) = G \frac{M(r)}{r^2}, \quad (2.89)$$

where $M(r)$ is the mass enclosed within the radius r .

Assuming hydrostatic pressure is sustained by elasticity. Here, the bulk modulus K represents the response of density to an infinitesimal change in pressure, satisfying $K = \rho \frac{dP}{d\rho}$. Thus, the pressure gradient becomes:

$$\frac{dP}{dr} = \frac{dP}{d\rho} \frac{d\rho}{dr} = \frac{K}{\rho} \frac{d\rho}{dr}. \quad (2.90)$$

^{note 11)} From these two equations, we obtain the Adams–Williamson equation: ^{(5), (8), (10)}

$$\frac{d\rho}{dr} = -\frac{\rho}{c_s^2} g(r). \quad (2.91)$$

Here, we define a density scale height H_ρ , a parameter frequently used to describe atmospheric structure, as:

$$H_\rho = \left(-\frac{1}{\rho} \frac{d\rho}{dr} \right)^{-1}. \quad (2.92)$$

^{note 10)} $U = \int \mathbf{r} \cdot \nabla P dV$ takes $\nabla \cdot \mathbf{r} = 3$ and uses Green's theorem. Ω is evaluated by bringing thin spherical shells from infinity to form a sphere, leading to $\Omega = -\int_0^M G M(r)/r dM(R)$.

^{note 11)} Here, since the bulk modulus is estimated from seismic wave velocities, we assume macroscopic deformation is adiabatic due to being sufficiently fast. If the temperature structure deviates from the adiabatic gradient, thermal expansion must be considered to balance static pressure. Moreover, effects from phase transitions or chemical changes require separate consideration.

It represents the thickness over which density decreases by a factor of $1/e$. Rearranging the equation yields:

$$c_s = \sqrt{H_\rho(r)g(r)}. \quad (2.93)$$

This formula corresponds directly to the earlier dimensional analysis. Our previous estimation using the Earth's radius R_e implicitly approximated the density scale height H_ρ with R_e . In the lower mantle, for instance, H_ρ is roughly 10^4 km, demonstrating that this is not a bad approximation.

2.10.3 Equation of state: Estimation of Earth's seismic wave velocity structure

To estimate a planet's seismic wave velocity structure, we need the relationship between density ρ and seismic wave velocity c_s (equation of state). Given the equation of state, we can estimate these structures by solving a coupled system of hydrostatic equilibrium, gravity, and the equation of state. First, we consider Earth, then expand to general planets.

Earth: Birch's law

Imagine learning about Earth's mass distribution without any prior information on its seismic wave velocities. Determining elastic constants requires learning the pressure-density relationship. For gases, the ideal gas law is well known; for solids, it is more complicated. Empirically, the variation of Earth's rock seismic velocities can be described as a function of density. Known as Birch's law,^{(2), (8)} it states:

$$c_s \approx a + b\rho. \quad (2.94)$$

Thus, seismic wave velocities can be described as a linear function of density^{note 12)}. Let us examine if this contradicts the Adams–Williamson equation for the real Earth.

Here, we assume $c_s = -2 \times 10^3 + \rho$ [kg/m³] in the inner and outer core, and $c_s = -1.75 \times 10^3 + 2.36\rho$ [kg/m³]⁽⁸⁾ in the mantle. Furthermore, we assume a density jump of 4.1424×10^3 [kg/m³] at the core–mantle boundary. Earth's total mass 5.972×10^{24} kg is assumed to be known. By coupling the Adams–Williamson equation and gravity formulation, we perform numerical integration from Earth's center toward the surface. By finding the distribution that yields exactly the total mass upon reaching the surface, we can estimate both the density and the bulk sound velocity.

The lower mantle reveals a remarkably accurate prediction, meaning it can be well represented by the adiabatic compression of a uniform material. On the other hand, estimations drastically diverge from actual PREM values in the upper mantle and crust. This occurs because the shallow regions deviate significantly from an adiabatic temperature gradient and because the compositional difference between the crust and the upper mantle renders our assumed Birch's parameters invalid.

Because Earth's deep interior consists of much heavier material (the core being predominantly iron), $g(r)$ in the lower mantle stays roughly constant (around the surface value), a behavior

^{note 12)}Strictly speaking, this approximates the equation of state using a first-order perturbation with respect to density around a certain reference value.

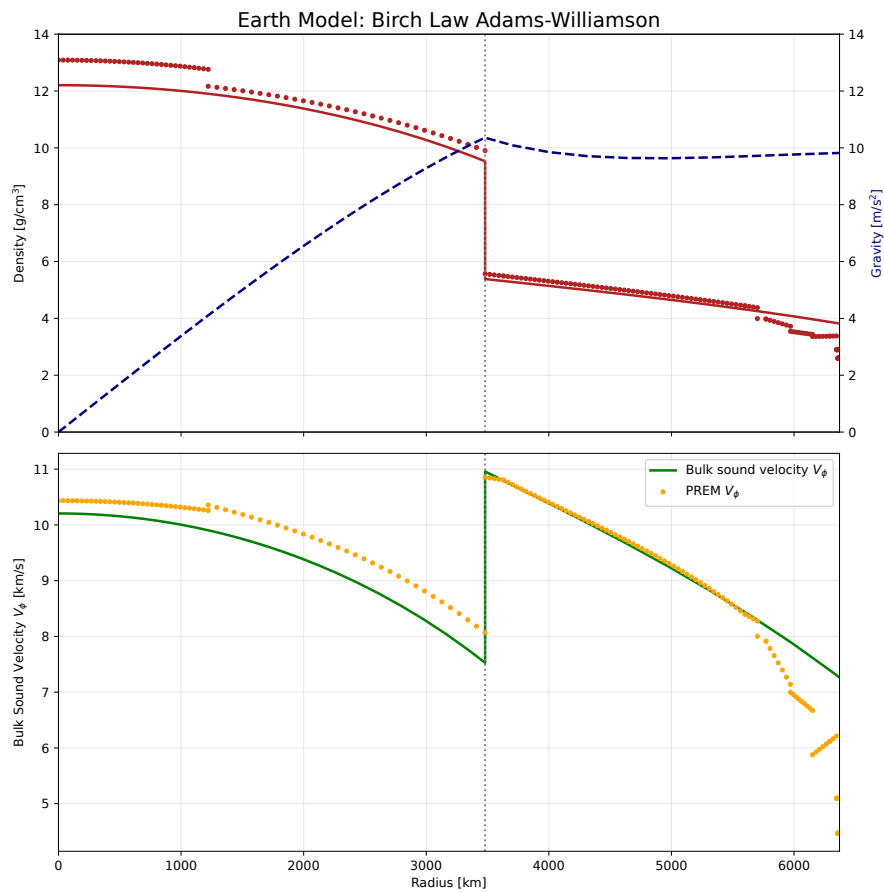


Figure 2.10: Estimated density structure and bulk sound velocity. PREM values are shown for reference. Birch's law was assumed, and the Adams–Williamson equation alongside the gravity profile was integrated outward from Earth's center.

beautifully reproduced by the computed model. Until seismic observations were made by the InSight mission, such density-based deductions⁽⁶⁾ were among the very few ways to estimate seismic wave velocity structures.

2.10.4 Seismic velocities in other planets: Planetary free oscillations

In the case of Earth, the Adams–Williamson equation has historically been used to estimate density structures from seismic wave velocity structures, since the latter is far more constrained than the former. When performing seismology on other planets, the situation is reversed. For many planets, although gravity can be accurately restrained through remote sensing, no empirical seismic observations exist. Estimating seismic wave-velocity structures by positing density bounds is extremely critical when initiating seismology on a new planet.

Here, let us consider an elastic free oscillation of a planet. As estimated earlier, if the planet's radius is R_p , its aggregate seismic wave velocity is bounded by $c_s = \sqrt{gR_p}$. Letting the average density be $\bar{\rho}$, gravitational acceleration g is given by:

$$g = \frac{4\pi}{3} G \bar{\rho} R_p. \quad (2.95)$$

Substituting this yields:

$$c_s = \sqrt{\frac{4\pi}{3} G \bar{\rho} R_p}. \quad (2.96)$$

We can estimate the fundamental eigenperiod T of elastic free oscillation as:

$$T = \frac{2\pi R_p}{c_s} = \frac{2\pi}{\sqrt{\frac{4\pi}{3} G \bar{\rho}}} = \sqrt{\frac{3\pi}{G \bar{\rho}}}. \quad (2.97)$$

The average density is roughly $5 \times 10^3 \text{ kg/m}^3$ for terrestrial solid planets, and $1 \times 10^3 \text{ kg/m}^3$ for gas giants and the Sun. Thus, the fundamental period of elastic free oscillations spans approximately 1000–5000 seconds. From this, we realize that while planets might differ radically in size, their global elastic oscillations share roughly the same temporal scale.

Planet / Composition	Density
Mercury	$5.43 \times 10^3 \text{ kg/m}^3$
Earth	$5.51 \times 10^3 \text{ kg/m}^3$
Venus	$5.24 \times 10^3 \text{ kg/m}^3$
Mars	$3.93 \times 10^3 \text{ kg/m}^3$
Jupiter	$1.33 \times 10^3 \text{ kg/m}^3$
Saturn	$0.69 \times 10^3 \text{ kg/m}^3$
Sun	$1.41 \times 10^3 \text{ kg/m}^3$

Table 2.1: Average density of the Sun and the planets.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

2.10.5 Power-law equation of state: Towards an understanding of generalized planets

In this subsection, we look into generalized equations of state to evaluate various types of planets, including gas giants.

A generalized equation of state relating pressure P and density ρ is posited via a power law:

$$P - P_0 \propto \left(\frac{\rho}{\rho_0} \right)^\gamma = \left(\frac{\rho}{\rho_0} \right)^{1+\frac{1}{n}} \quad (2.98)$$

$$\left(\frac{c_s}{c_0} \right)^2 \propto \left(\frac{\rho}{\rho_0} \right)^{\gamma-1} = \left(\frac{\rho}{\rho_0} \right)^{\frac{1}{n}}. \quad (2.99)$$

Here, n represents the polytropic index. Larger values of n imply a gaseous behavior, whereas lower values of n indicate solid behaviors. ρ_0 , c_0 , and P_0 correspond to surface reference values.

Isothermal atmosphere: $n = \infty$

For $\gamma = 1$ ($n = \infty$), the equation of state is:

$$(c_s)^2 \propto \rho^{\gamma-1} = \text{constant}. \quad (2.100)$$

Since the speed of sound is proportional to the square root of temperature, this configuration proves an isothermal atmospheric structure.

Stability limit: $n = 3$

Through dimensional analysis, overall mass M and radius R subject to zero-density bounds follow:

$$M \propto R^{\frac{3-n}{1-n}}. \quad (2.101)$$

For simplicity, we define the radius R as the shell where the density is 0. Under $n = 3$, M ceases to depend on R . The existence of equivalent total masses at assorted radii implies that infinitesimally smaller sizes become more gravitationally stable. This triggers sudden instability, leading to rapid planetary deflation once compression begins.

Monatomic ideal gas: $n = 1.5$

For the adiabatic expansion of a monatomic ideal gas, $\gamma = 5/3$ ($n = 1.5$), which gives the equation of state:

$$(c_s)^2 \propto \rho^{\frac{1}{3}}. \quad (2.102)$$

Bullen's density model: $n = 1$

This model was deployed by Bullen to approximate the Earth as a single, uniform shell. From $M \propto R^{\frac{3-n}{1-n}}$, setting $n = 1$ enforces that the radius loses density dependence limitlessly. For $n < 1$, heavier masses yield smaller radii, whereas for $n > 1$, heavier masses swell the target radius. Hence, $n = 1$ partitions gaseous and solid behaviors. The coupled equation of state reads:

$$(c_s)^2 \propto \rho. \quad (2.103)$$

In this scenario, an analytical solution exists, modeling the density configuration as:

$$\rho(r) = \rho_c \frac{\sin \xi_0 r}{\xi_0 r}, \quad (2.104)$$

where $\xi_0 = \sqrt{4\pi G \rho} / c_s$.

Solid (rock) planetary model: $n = 1/3$

The equation of state corresponds to:

$$(c_s)^2 \propto \rho^3. \quad (2.105)$$

As observed from $c_s = \sqrt{g H_\rho}$, a softened surface velocity parameter c_0 reduces the density scale height H_ρ , localizing lower velocities exclusively along the outer surface layers.

§2.11 Comparison of Gravity and Elastic Terms

Let us estimate the orders of magnitude of the gravity and elastic terms in the equations of motion for a wave $s = e^{i(kx - \omega t)}$, where ω is the angular frequency and k is the wavenumber. The inertia term is $\rho \partial_t^2 s \approx -\rho \omega^2 s$. The elasticity term $\nabla \cdot \mathbf{T}$ is estimated as $k^2 \kappa s$, where κ is a typical elastic constant. The gravity term corresponds to the second term on the right side of Eq. (2.62), which relates to buoyancy and is estimated as $\rho k s g$. Here, g is gravity acceleration. A comparison between the gravity term and the elasticity term is given by,

$$\frac{\text{Gravity}}{\text{Elasticity}} \sim \frac{\rho g}{k \kappa} = \frac{\rho g \lambda}{4\pi \kappa} = \frac{\rho g \lambda c^2}{4\pi \kappa c^2} = \frac{g T}{4\pi c}. \quad (2.106)$$

where we used wavelength λ , period T , and typical seismic wave velocity c . For the gravity term to be significant, the time scale must be $T \sim \frac{4\pi c}{g}$. As estimated earlier, a typical seismic wave velocity c can be approximated as $\sqrt{g R_e}$ using the Earth's radius R_e . From this relation^{note 13)}, the

^{note 13)} Using $g = c^2 / R_e$.

period T where gravity and elasticity become comparable is:

$$T \sim \frac{4\pi c}{g} = \frac{4\pi R_e}{c} \sim \frac{4\pi \times 6400 \times 10^3}{10 \times 10^3} \sim 8000 \text{ s.} \quad (2.107)$$

This is consistent with the periods of the Earth's fundamental modes of longest period (free oscillations), demonstrating that we can safely neglect the gravity term for periods shorter than 100 s. For acoustic waves in the atmosphere (sound velocity ~ 340 m/s), the gravity term becomes comparable to the compressibility term at a period of about 200 s (acoustic-gravity waves).

Next, let us consider the effect of **self-gravity** (the perturbation of the gravitational potential ϕ^{E1}). This corresponds to the third term on the right side of Eq. (2.62). Its magnitude is on the order of $\rho \nabla \phi^{E1} \sim \rho k \phi^{E1} \sim \rho k \left(\frac{4\pi G \rho}{k^2} s \right) = 4\pi \frac{G \rho^2}{k} s$ (where we used Poisson's equation to convert ϕ^{E1}). Comparing this self-gravity term with the elasticity term yields:

$$\frac{\text{Self-gravity}}{\text{Elasticity}} \sim 4\pi \frac{\rho^2 G}{k^2 \kappa} = \frac{4\pi G \rho}{k^2 c^2} = \frac{3g}{R_e k^2 c^2}, \quad (2.108)$$

where we used the relation $4/3\pi\rho GR_e \sim g$. Using the typical velocity $c \sim \sqrt{gR_e}$, the period T at which self-gravity and elasticity become comparable is:

$$T \sim \frac{2\pi R_e}{\sqrt{3}c} \approx \frac{2\pi R_e}{c}. \quad (2.109)$$

This also indicates an order of magnitude of several thousand seconds.

At periods longer than 1000 s, the Coriolis force originating from Earth's rotation is not negligible. In such a case, seismic wave velocity depends on the propagation direction^{note 14}. Figure 2.11 shows the splitting of resonant peaks of ${}_0S_2$ due to the Coriolis force during the 2004 Sumatra–Andaman earthquake. The directivity breaks the reciprocity of the elastic Green's function, which is explained in the following chapter (see Dahlen and Tromp 1998 for details). Read Snieder *et al.* (2016)⁽⁹⁾ for details of the Coriolis effects.

^{note 14}) An example of the 1960 great Chilean earthquake can be found in The Feynman Lectures on Physics http://www.feynmanlectures.caltech.edu/I_51.html. We can imagine the atmosphere of Caltech in 1960'

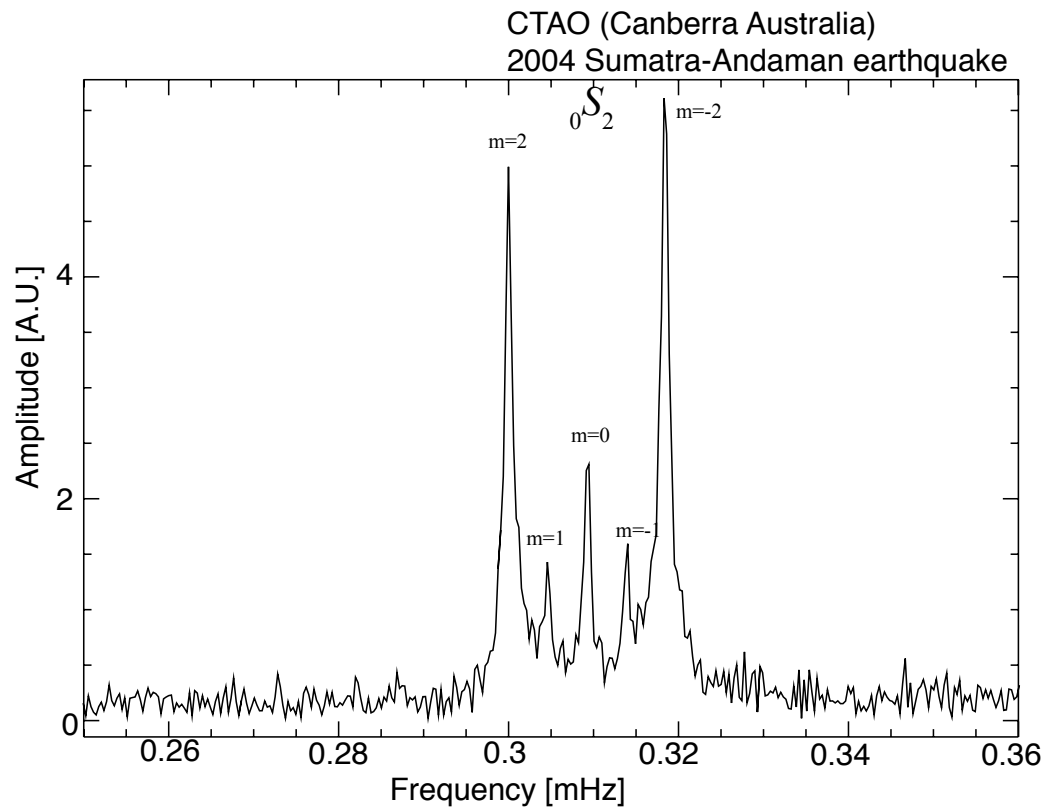


Figure 2.11: Coriolis splitting of the resonant peak of ${}_0S_2$ when the great Sumatra-Andaman earthquake in 2004.

Problem 2.9

Based on a comparison of the Coriolis term and elastic restoration force, estimate the period at which the Coriolis force becomes significant.

§2.12 Boundary conditions

When solving equations of motion, boundary conditions are indispensable. The Lagrangian description of the boundary conditions is natural. However, we do not account for the difference when we do not consider initial stress.

2.12.1 Solid-solid boundaries such as Moho and 660 km discontinuity

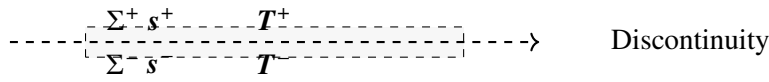


Figure 2.12: Schematic figure of a plane Σ^+ and Σ^- .

Let us consider a pair of closely placed planes Σ^+ and Σ^- encompassing the boundary. The equilibrium of forces and the continuity of displacement within this intermediate region yield the fundamental boundary conditions that must be satisfied across any internal discontinuity in an elastic body.

First, by requiring physical continuity, the continuity of displacement is imposed as $[s]_{\pm}^+ = \mathbf{0}$. This means that the material particles across the boundary remain continuously attached, without separating or interpenetrating each other, after deformation. Next, regarding mechanical interaction, the continuity of total traction must hold, leading to $[T^{L1} \cdot \hat{n}]_{\pm}^+ = \mathbf{0}$. Following the principle of action and reaction, the force per unit area transmitted through the normal vector $\hat{n}$ must perfectly balance across the boundary. These conditions are the most fundamental constraints determining how elastic waves reflect and transmit at discontinuous boundaries.

2.12.2 Solid-fluid boundaries such as ocean floor and core-mantle boundary

At solid-fluid boundaries (such as the seafloor or the core-mantle boundary), the formulation of boundary conditions is of critical importance. In particular, when the fluid is treated as an ideal inviscid fluid, free slip along the solid wall is permitted, which fundamentally distinguishes it from a pure solid-solid boundary. At this interface, the continuity of displacement is first imposed as $[s \cdot \hat{n}]_{\pm}^+ = 0$. This ensures that the normal component of displacement is identical on both sides, preventing any cavitation or overlap at the boundary. However, a discontinuity in the components parallel to the boundary, or **slip**, is permitted. This horizontal discontinuity allows the existence of special waves where energy is strictly localized near the boundary. The **Stoneley mode**, which will be detailed in the next chapter, is a prime example of this; its energy is concentrated at the seafloor and the CMB, making it notoriously difficult to detect from surface observation networks.

Next, the mechanical balance is defined by the continuity of total traction, denoted as $[\mathbf{T}^{L1} \cdot \hat{\mathbf{n}}]_{-}^{+} = \hat{\mathbf{n}}[\hat{\mathbf{n}} \cdot \mathbf{T}^{L1} \cdot \hat{\mathbf{n}}]_{-}^{+} = \mathbf{0}$. Crucially, because shear stress is always zero in the fluid, the traction transmitted across the boundary is strictly limited to the pressure component in the normal direction. Note that when dealing with systems dominated by massive initial stress or hydrostatic pressure, such as deep inside the Earth, these contributions cannot be neglected and require extra care in formulation.⁽¹⁾

2.12.3 Continuity of gravity potential for all boundaries

When addressing vibration problems in elastic bodies involving self-gravity, boundary conditions require meticulous formulation. Inherently, elastic deformations are most naturally represented using the Lagrangian description, which tracks material points. Contrarily, describing fields such as gravitational potentials naturally necessitates the Eulerian description, which is referenced to fixed spatial points. In typical problems where the initial stress is negligible, distinguishing between these two descriptions is rarely necessary. However, when accommodating self-gravitation involving the perturbation of the gravitational potential ϕ , the discrepancy between both definitions directly infuses into the structure of boundary conditions. Particularly at discontinuity surfaces, boundary conditions are consolidated utilizing the Eulerian perturbation of potential ϕ^{E1} as follows.

First, because the gravitational potential itself must remain smoothly continuous across the discontinuity surface, the continuity condition for the potential acts as $[\phi^{E1}]_{-}^{+} = 0$. Next, concerning the continuity condition for the vertical component of gravity, the alteration in mass geometry (the effect of the density discontinuity) accompanying boundary displacement must be evaluated. This manifests as a combination of the jump in the potential's gradient and terms dependent on both the displacement vector $\mathbf{s}$ and the background density ρ_0 , formulated as $[\mathbf{n} \cdot \nabla \phi^{E1} + 4\pi G \rho_0 \mathbf{n} \cdot \mathbf{s}]_{-}^{+} = 0$. Detailed inspections corresponding to the rigorous separation of these descriptions are examined thoroughly later in subsequent chapters.

§2.13 Bibliography

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